

DRUMALBIN, United KingdomWMO: **031550**Lat: **55.6272N**Lon: **3.7361W**Elev: **245**StdP: **98.42**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.3	-3.5	-6.6	2.2	-3.8	-5.3	2.5	-2.1	19.1	5.8	17.7	5.6	1.6	100	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	21.7	16.6	20.0	15.5	18.4	14.8	17.5	20.5	16.4	18.9	15.5	17.6	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.9	18.7	15.4	11.2	17.5	14.5	10.6	16.7	50.1	20.4	46.9	19.0	44.2	17.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	13.7	12.1	DB	-7.9	25.1	2.4	1.9	-9.6	26.5	-11.0	27.6	-12.4	28.7	-14.1	30.1	(4)
(5)			WB	-8.1	19.6	2.4	1.8	-9.8	20.9	-11.2	22.0	-12.5	23.0	-14.2	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	3.0	3.1	4.4	6.8	9.3	12.0	13.8	13.5	11.7	8.7	5.4	3.2
	DBStd	4.81	2.99	2.79	2.73	2.69	2.90	2.40	2.24	2.03	2.30	2.71	3.05	3.70
	HDD10.0	1156	217	194	172	101	48	7	1	1	9	57	139	211
	HDD18.3	3799	475	427	431	346	280	191	143	152	199	299	388	468
	CDD10.0	399	0	0	0	5	26	66	117	108	60	16	2	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	21	0	0	0	0	1	4	11	4	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	1	0	0	0	0	0
	WSAvg	5.5	6.4	6.4	6.2	5.6	5.3	4.9	4.4	4.7	5.3	5.7	5.7	5.9
(15) Precipitation	PrecAvg	1187	135	103	88	68	74	71	88	91	89	127	123	132
	PrecMax	1445	249	266	180	107	130	171	149	228	153	218	254	311
	PrecMin	875	17	18	23	15	16	25	28	18	17	34	50	31
	PrecStd	156	53	63	34	22	34	34	32	51	35	55	53	67
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	10.8	13.8	18.5	21.7	23.5	24.9	23.1	20.4	16.0	13.3	11.7
		MCWB	9.6	9.4	9.0	12.8	15.6	16.8	18.0	18.2	16.0	13.9	12.3	10.8
	2%	DB	9.4	9.3	10.8	14.9	19.2	20.4	22.0	20.0	18.2	14.6	11.9	10.4
		MCWB	8.6	8.2	8.5	10.4	13.8	15.8	17.3	16.2	14.7	13.1	11.0	9.6
	5%	DB	8.4	8.4	9.5	12.7	16.4	18.3	20.0	18.4	16.7	13.7	10.9	9.4
		MCWB	7.6	7.4	7.7	9.5	12.4	14.6	15.8	15.2	14.2	12.4	9.9	8.6
	10%	DB	7.5	7.4	8.5	11.1	14.2	16.6	18.1	17.2	15.5	12.7	9.9	8.3
		MCWB	6.6	6.3	7.0	8.6	10.9	13.4	14.9	14.5	13.4	11.4	8.9	7.5
	0.4%	WB	10.0	10.0	10.2	13.0	16.7	18.2	19.2	18.5	17.2	14.6	12.5	10.9
		MCDB	10.3	10.5	12.4	17.2	20.9	21.9	23.6	21.6	19.4	15.2	13.0	11.4
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	8.8	8.5	9.1	11.3	14.4	16.5	17.7	16.9	15.7	13.5	11.1	9.7
		MCDB	9.3	9.2	10.3	14.1	18.1	19.6	21.1	18.9	17.1	14.3	11.7	10.3
	5%	WB	7.7	7.6	8.1	10.0	12.8	15.0	16.4	16.0	14.7	12.6	10.2	8.7
		MCDB	8.3	8.3	9.1	12.1	15.8	17.6	19.1	17.8	16.1	13.5	10.8	9.4
	10%	WB	6.7	6.5	7.2	9.0	11.5	13.8	15.3	15.1	13.7	11.6	9.0	7.6
		MCDB	7.4	7.3	8.3	10.8	13.8	16.0	17.6	16.7	15.2	12.6	9.8	8.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.9	5.9	7.1	7.9	7.5	7.4	7.0	6.4	5.5	4.8	4.5
		MCDBR	4.9	5.4	7.0	9.7	11.4	10.7	10.7	8.7	8.0	5.9	5.2	5.3
		MCWBR	5.0	5.0	5.3	6.5	7.1	6.8	6.4	5.5	5.3	4.9	5.0	5.3
	5% WB	MCDBR	4.9	5.2	6.5	8.4	10.3	9.9	9.3	7.7	6.8	5.5	5.3	5.3
		MCWBR	5.0	5.1	5.3	6.0	6.8	6.9	6.1	5.5	5.3	5.0	5.3	5.4
(40) Clear-Sky Solar Irradiance	taub		0.316	0.328	0.354	0.373	0.375	0.372	0.373	0.358	0.347	0.331	0.308	0.299
	taud		2.236	2.344	2.308	2.285	2.321	2.354	2.354	2.407	2.447	2.436	2.325	2.212
	Ebn at Noon		582	741	807	846	866	872	862	854	813	732	598	505
	Edh at Noon		57	75	97	114	117	115	113	100	84	67	52	46
(44) All-Sky Solar Radiation	RadAvg		0.53	1.17	2.22	3.61	4.70	4.57	4.40	3.56	2.53	1.39	0.68	0.39
	RadStd		0.07	0.12	0.22	0.37	0.37	0.45	0.47	0.28	0.20	0.17	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page